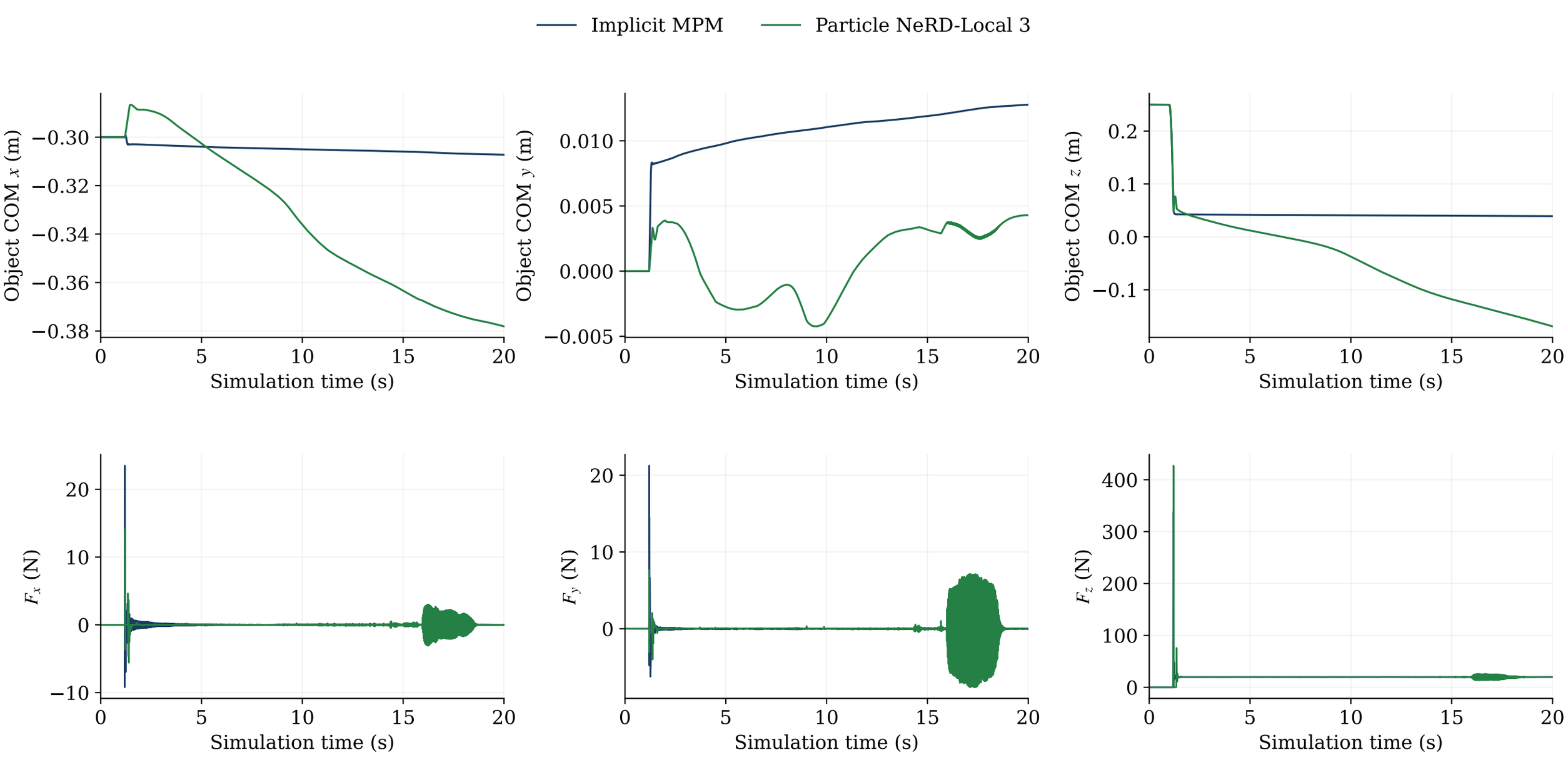


Figure 10: Sphere, one 20 s fully dynamic rollout per method from identical initial states. Top row: object center-of-mass x, y, z in world coordinates. Bottom row: raw material-on-object Fx, Fy, Fz; direct rigid-tray forces are excluded. The upward holding force is removed at 1 s. Each method evolves its own object and particles; NeRD force and torque feed back through the next 5 ms coupling exchange. All 200 Hz samples are shown without smoothing. The paired video renders these same saved states at 50 fps.